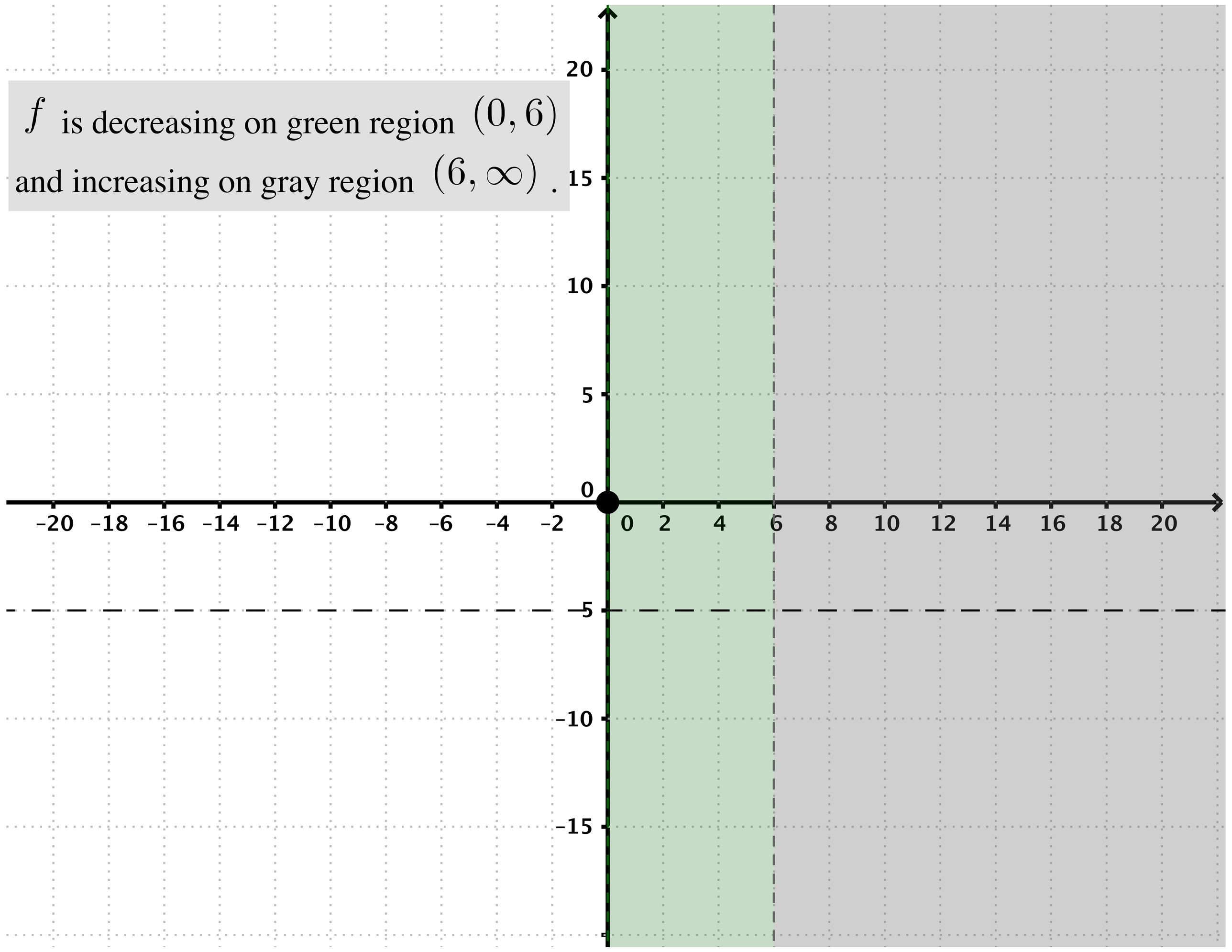f is decreasing on green region $(0, 6)$ and increasing on gray region $(6, \infty)$.